7 **(a)** Fig. 7.1 shows a 12.0 V cell with negligible internal resistance connected in series with a uniform resistance wire XY of length 6.0 cm.

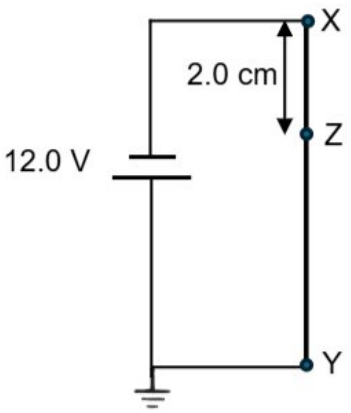


Fig. 7.1

(i) Point Z is 2.0 cm away from point X. Determine the potential at point Z.

potential at point Z = ... V [3]

(ii) Show that the resistance of a wire is given by

$$R = \frac{\rho D L^2}{m}$$

where ρ is the resistivity of the wire, D is the density of the wire, L is the length of the wire and m is the mass of the wire.

[2]

(iii) A student suggests that, if the length of wire XY is doubled with same cross sectional area, the resistance of wire XY increases by a factor of four.

Explain whether the student is correct.

.....

.....

.....

..... [2]

(b) **(i)** The *I*-*V* characteristic graph for a NTC thermistor is shown in Fig. 7.2.

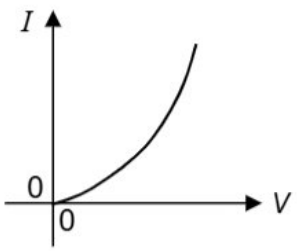


Fig. 7.2

Explain how the resistance of a NTC thermistor changes with increase in potential difference in terms of the number density of charge carriers.

.....

.....

.....

.....

.....

.....

..... [4]

(ii) The NTC thermistor is connected in a circuit with a cell, a resistor and a light emitting diode (LED) so that the LED emits light at high temperature.

Draw a circuit diagram showing how the components are connected and explain how the LED is able to emit light at high temperature.

.....

.....

.....

..... [4]

(iii) Explain why the resistor is needed in the circuit in part **b(ii)**.

.....

..... [1]

(c) An initially uncharged capacitor of capacitance 2.0 μF is charged by a battery of e.m.f. 5.0 V through a resistor of resistance 3.0 MΩ, as shown in Fig. 7.3. The switch is closed at time $t = 0$.

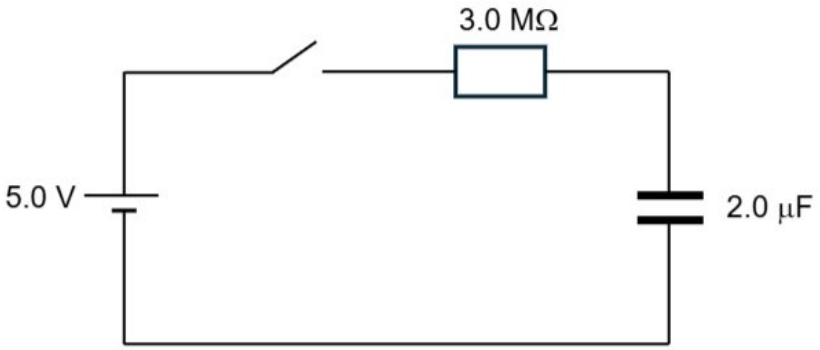


Fig. 7.3

(i) In Fig. 7.4, sketch the variation with time t of potential difference V across the capacitor, until it is fully charged.

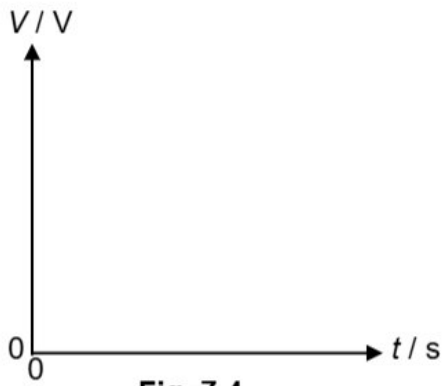


Fig. 7.4

[2]

(ii) Determine the time taken to charge the capacitor to 2.0 V.

time = ... s [2]